Diabetes Screening Protocols for Community Medicine Posting

MBBS Students - Community Health Camp Guidelines

Introduction

Community-based diabetes screening is a core competency for MBBS students in community medicine posting. This protocol ensures standardized, evidence-based screening while maintaining ethical and cultural sensitivity.

Screening Eligibility Criteria

Who to Screen:

☑ **Risk Group A (High Priority):**

- Age ≥ 35 years
- BMI ≥ 23 kg/m² (Asian criteria)
- Family history of diabetes in first-degree relative
- Previous gestational diabetes
- Hypertension (BP ≥ 140/90 mmHg)

Risk Group B (Medium Priority):

- Age 30-34 years
- Overweight (BMI 23-25 kg/m²)
- Limited physical activity
- Urban residents with sedentary lifestyle
- Previous abnormal blood glucose

Universal Screening:

- **All adults aged ≥ 30 years** in high-risk communities
- **Any person with suspicious symptoms**
- **Annual screening** for all diabetics

Screening Tools and Equipment

Essential Equipment:

- 💧 **Glucometer** (validated, calibrated)
- 📏 **Test strips** (batch checked for validity)
- 💉 **Lancets** (single-use, safety lancets preferred)
- 🧴 **Alcohol swabs** and cotton

- 🧤 **Disposable gloves**
- 📋 **Screening forms** and ID cards

Optional Equipment:

- 📏 **Measuring tape** for BMI
- ⚖️ **Weighing scale**
- 📐 **Stadiometer** for height
- 🩺 **BP apparatus**
- 📱 **Mobile app** for data entry

Screening Process (Step-by-Step)

Phase 1: Camp Setup (30 minutes)

1. **Location:** Community center, school, or PHC with adequate space
2. **Team Composition:** 1 faculty + 2-3 students + 1 lab technician
3. **Safety:** Handwashing stations, biohazard disposal, emergency kit
4. **Materials:** Consent forms, privacy screens, educational posters

Phase 2: Community Mobilization (1 hour)

1. **Announcement:** Loudspeaker, posters, local leaders
2. **Education:** Pre-camp awareness session
3. **Registration:** Token system to manage crowds
4. **Privacy:** Separate areas for men/women if cultural preference

Phase 3: Individual Screening (5-7 minutes per person)

Step 1: Initial Assessment (2 minutes)

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- Consent: "I have read and understood the screening process"
- Basic details: Name, age, contact, address
- Risk factors: Family history, symptoms, medications
- Anthropometry: Height, weight, BMI calculation
- BP measurement: Two readings, average taken
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Step 2: Blood Glucose Testing (2 minutes)

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- Explain procedure: "Small finger prick, mild discomfort"
- Select site: Lateral aspect of ring finger
- Clean with alcohol, allow to dry
- Lance quickly, collect first drop
- Apply to test strip, read result within time window
- Record fasting/RBS status
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Step 3: Results Counseling (3-5 minutes)

Normal Results (RBS < 140 mg/dL):

- Explain normal vs abnormal
- Provide prevention advice
- Schedule follow-up screening

Abnormal Results (RBS 140-199 mg/dL):

- Explain pre-diabetes condition
- Provide dietary and lifestyle advice
- Refer for confirmatory OGTT
- Follow-up in 3-6 months

High Results (RBS ≥ 200 mg/dL):

- Immediate medical attention
- Temporary dietary advice
- Refer to nearest PHC
- Home glucometer education if possible
- Family counseling

Clinical Decision Making

Interpretation Guidelines:

RBS Level	Interpretation	Action	Referral
< 140 mg/dL	Normal	Lifestyle advice	1 year follow-up
140-199 mg/dL	Impaired glucose tolerance	Lifestyle + metformin?	OGTT booking
200-300 mg/dL	Diabetes likely	Immediate care	Same day PHC
> 300 mg/dL	Severe hyperglycemia	Emergency care	Hospital admission

Special Considerations:

- **Fasting status:** RBS vs FBS interpretation differences
- **Pregnancy:** Specific cut-offs for gestational screening
- **Children:** Pediatric protocols if encountered
- **Elderly:** Consider frailty and comorbidities

Data Management & Reporting

Individual Records:

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Patient ID: Unique identifier
Demographics: Age, sex, address
Measurements: BP, BMI, RBS results
Risk factors: Family history, symptoms
Outcome: Normal/abnormal, referrals made
Counseling: Topics covered, materials provided
Follow-up: Date and contact information
```

Camp Summary:

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Total screened: XX
Normal: XX (XX%)
Pre-diabetes: XX (XX%)
Diabetes: XX (XX%)
Referrals: XX
Materials distributed: XX
Community feedback: Notes
```

Data Quality:

- **Real-time entry:** No backlog of data
- **Accuracy checks:** 10% random verification
- **Confidentiality:** No identifiers in public reports
- **Storage:** Secure digital + physical backup

Infection Control & Safety

During Screening:

- Hand hygiene between patients
- Single-use lancets, gloves, swabs
- Biohazard waste segregation
- AAA (Airway, Breathing, Circulation) kit ready for hypoglycemic reactions

Equipment Maintenance:

- Glucometer calibration daily
- Strip storage as per manufacturer
- Emergency kit check daily
- Cold chain maintenance for reagents

Environmental Safety:

- Shade and ventilation
- Clean water availability
- Emergency exit routes

- Insect/mosquito protection

Community Engagement

Pre-Camp:

- Local leader involvement
- Religious/cultural considerations
- Language-appropriate materials
- Transportation/logistics planning

During Camp:

- Respect local customs
- Flexible timing for prayers/meals
- Family members welcome
- Positive reinforcement

Post-Camp:

- Follow-up mechanisms
- Community feedback collection
- Success stories sharing
- Continuous improvement

Quality Assurance

Process Indicators:

- Average screening time: < 8 minutes
- Consent rate: > 95%
- Data completeness: > 90%
- Patient satisfaction: > 80%

Outcome Indicators:

- Appropriate referral rates
- Patient follow-up rates
- Community impact assessment
- Student learning evaluation

Troubleshooting Guide

Equipment Issues:

- **Glucometer error:** Check strips, calibration, cleaning
- **Low battery:** Keep charged spare available

- **Strip expired:** Check expiry before camp

Patient Issues:

- **Fear of needle:** Distraction, start with less anxious
- **Multiple languages:** Translator assistance, pictorial aids
- **Emergency:** Have hypoglycemia management protocol ready

Logistical Issues:

- **Crowd management:** Token system, breaks for staff
- **Weather:** Indoor backup plan
- **Power failure:** Manual recording system

CBME Competencies Addressed

Patient Care (PC):

- PC1: Community screening protocols
- PC4: Complication risk assessment

Systems-Based Practice (SBP):

- SBP1: Community health system navigation
- SBP2: Resource utilization in camps
- SBP3: Quality improvement in screening

Professionalism (PRO):

- PRO1: Ethical consent and data handling
- PRO2: Cultural competence in camps

Student Self-Assessment Checklist

Screening Camp Competency Checklist:

Preparation: □ Equipment checklist completed □ Emergency protocols reviewed □ Cultural considerations assessed □ Team roles assigned

During Camp: □ All patients handled with respect □ Accurate results interpretation □ Appropriate counseling provided □ Referrals made correctly □ Data recorded completely

Post-Camp: □ Equipment cleaned and stored □ Data analysis completed □ Report submitted timely □ Learning reflection documented

Faculty Evaluation Required: YES/NO

SOP References

- **Indian Council of Medical Research (ICMR)** Guidelines
- **National Programme for Prevention and Control of NCDs** (NP-NCD)
- **WHO Package of Essential Non-communicable diseases (PEN)**
- **RSSDI Diabetes Screening Guidelines**

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